Task 1

```
Original_image_size = 200 * 200
Resized_image_size = 100 * 100


scale_x = 100 / 200
scale_y = 100 / 200


x_min, y_min = 40 * scale_x , 60 * scale_y
x_max, y_max = 120 * scale_x, 160 * scale_y

print(f"x_min: {x_min}, y_min: {y_min}")
print(f"x_max: {x_max}, y_max: {y_max}")
```

```
x_min: 20.0, y_min: 30.0
x_max: 60.0, y_max: 80.0
```

Task 2

```
import os
import zipfile
from pathlib import Path

import cv2
import matplotlib.pyplot as plt
import tensorflow as tf

from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense
from tensorflow.keras.models import Model

from google.colab import files
```

```
from pathlib import Path
import cv2
import matplotlib.pyplot as plt

test_path = Path("Covid19-dataset/train/Covid")

images = list(test_path.glob("*.jpeg"))[:3]
```

```
base_model = MobileNetV2(
    weights="imagenet",
    include_top=False,
    input_shape=(100,100,3)
)
base_model.trainable = False
```

```
/tmp/ipykernel_2439/485835247.py:1: UserWarning: `input_shape` is undefined or non-square, or `rows` is not in [96, 128, 160, 19
  base_model = MobileNetV2(
```

```
x = base_model.output
x = GlobalAveragePooling2D()(x)
output = Dense(3, activation="softmax")(x)

model = Model(inputs=base_model.input,
              outputs=output)
model.compile(
    optimizer="adam",
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
model.summary()
```

```
fig, axes = plt.subplots(3, 1, figsize=(5,8))

for i, file in enumerate(images):

    image = cv2.imread(str(file))
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # Resize image
    image = cv2.resize(image, (100,100))

    # Scaled bounding box
    x_min = 20
    y_min = 30
    x_max = 60
    y_max = 80

    cv2.rectangle(
        image,
        (x_min, y_min),
        (x_max, y_max),
        (255, 0, 0),
        1
    )

    axes[i].imshow(image)
    axes[i].set_title(f"Image {i+1}")
    axes[i].axis("off")

plt.tight_layout()
plt.show()
```

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